Optical character recognition (Wikipedia)

Optical character recognition (OCR) or optical character reader is the electronic or mechanical conversion of images of typed, handwritten or printed text into machine-encoded text, whether from a scanned document, a photo of a document, a scene photo (for example the text on signs and billboards in a landscape photo) or from subtitle text superimposed on an image (for example: from a television broadcast).

Widely used as a form of data entry from printed paper data records – whether passport documents, invoices, bank statements, computerized receipts, business cards, mail, printed data, or any suitable documentation – it is a common method of digitizing printed texts so that they can be electronically edited, searched, stored more compactly, displayed online, and used in machine processes such as cognitive computing, machine translation, (extracted) text-to-speech, key data and text mining. OCR is a field of research in pattern recognition, artificial intelligence and computer vision.

Early versions needed to be trained with images of each character, and worked on one font at a time. Advanced systems capable of producing a high degree of accuracy for most fonts are now common, and with support for a variety of image file format inputs. Some systems are capable of reproducing formatted output that closely approximates the original page including images, columns, and other non-textual components.

History

Early optical character recognition may be traced to technologies involving telegraphy and creating reading devices for the blind. In 1915, Hyman Eli Goldberg was granted a patent for a device that read perforated or conductive-ink characters and converted them into standard telegraph code (US Patent 1165663). Concurrently, Edmund Fournier d'Albe developed the Optophone, a handheld scanner that when moved across a printed page, produced tones that corresponded to specific letters or characters.

In the late 1920s and into the 1930s, Emanuel Goldberg developed what he called a "Statistical Machine" for searching microfilm archives using an optical code recognition system. In 1931, he was granted US Patent number 1,838,389 for the invention.

In 1974, Ray Kurzweil started the company Kurzweil Computer Products, Inc. and continued development of omni-font OCR, which could recognize text printed in virtually any font. (Kurzweil is often credited with inventing omni-font OCR, but it was in use by companies, including CompuScan, in the late 1960s and 1970s.) Kurzweil used the technology to create a reading machine for blind people to have a computer read text to them out loud. The device included a CCD-type flatbed scanner and a text-to-speech synthesizer. On January 13, 1976, the finished product was unveiled during a widely reported news conference headed by Kurzweil and the leaders of the National Federation of the Blind. In 1978, Kurzweil Computer Products began selling a commercial version of the optical character recognition computer program. LexisNexis was one of the first customers, and bought the program to upload legal paper and news documents onto its nascent online databases. Two years later, Kurzweil sold his company to Xerox, which eventually spun it off as Scansoft, which merged with Nuance Communications.